

PDF Analysis: Neural Network Synthetic Local Volatility Model

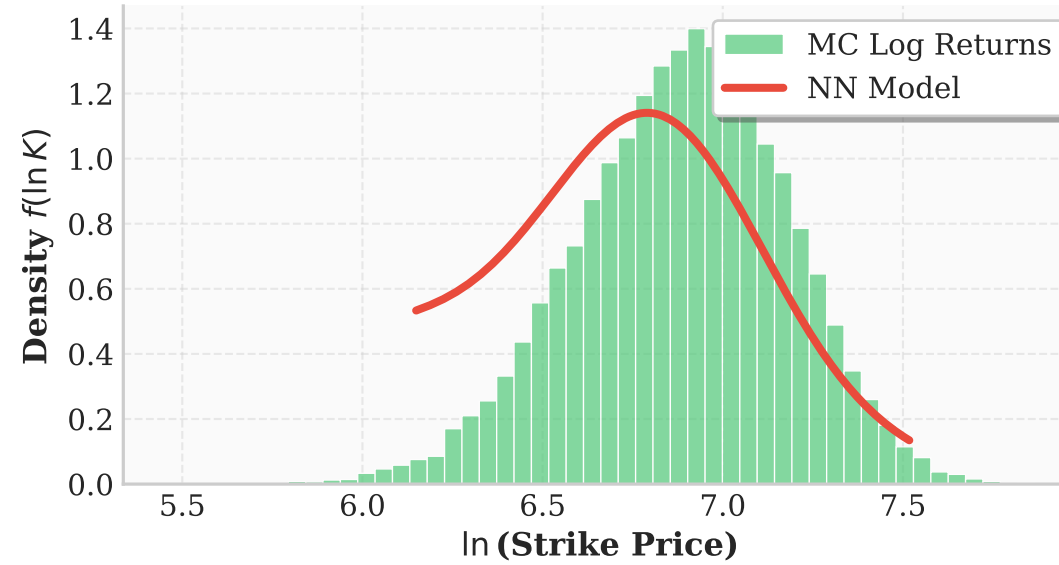
Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S)S_t dW_t$

Retrained (K0 BC penalty, K_min=100)

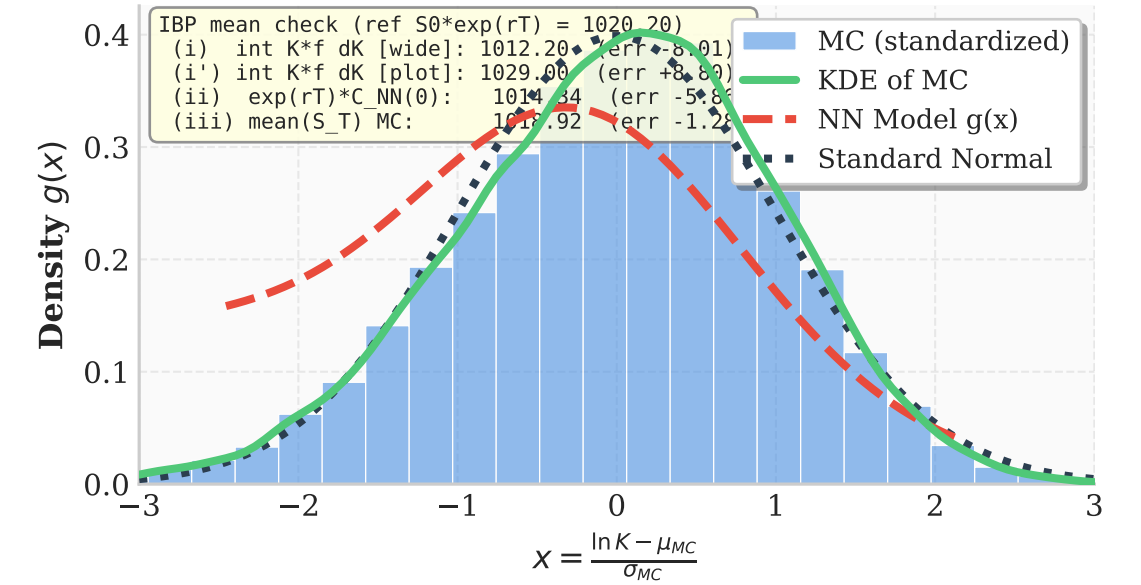
Strike Distribution (T = 0.50)



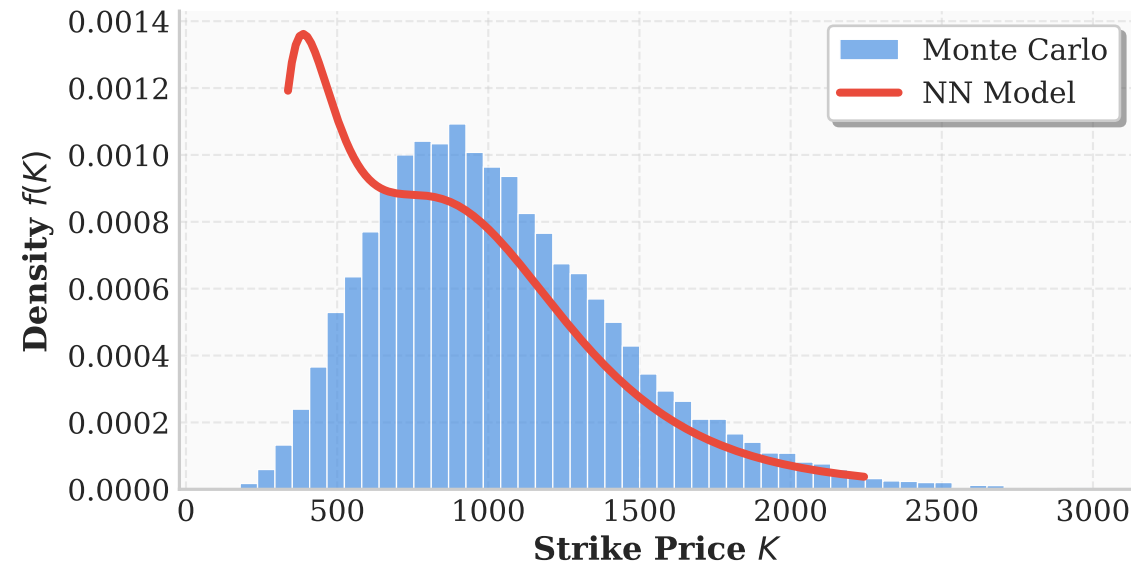
Log-Normal Distribution (T = 0.50)



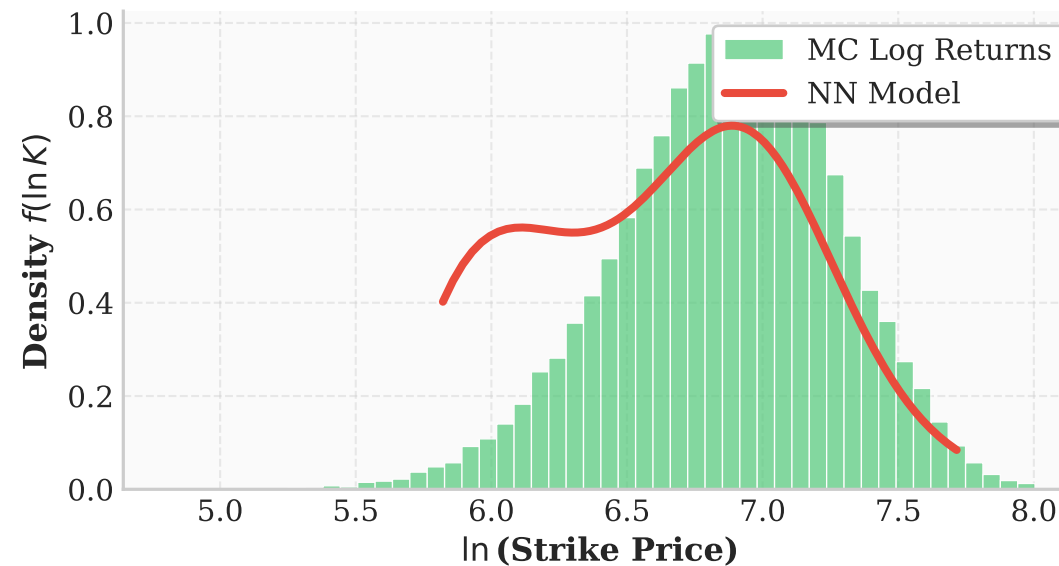
Gaussian Space (T = 0.50)
MC Standardization: $\mu_{MC}=6.88, \sigma_{MC}=0.29 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.76, \sigma_{model}=0.32$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.26, \text{ExKurt}=0.11$



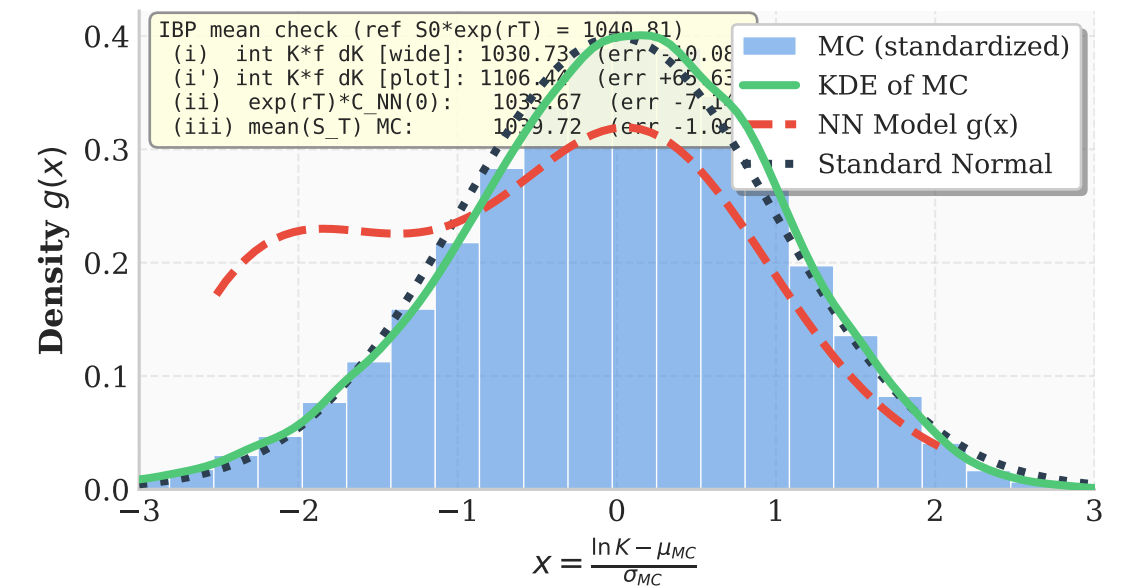
Strike Distribution (T = 1.00)



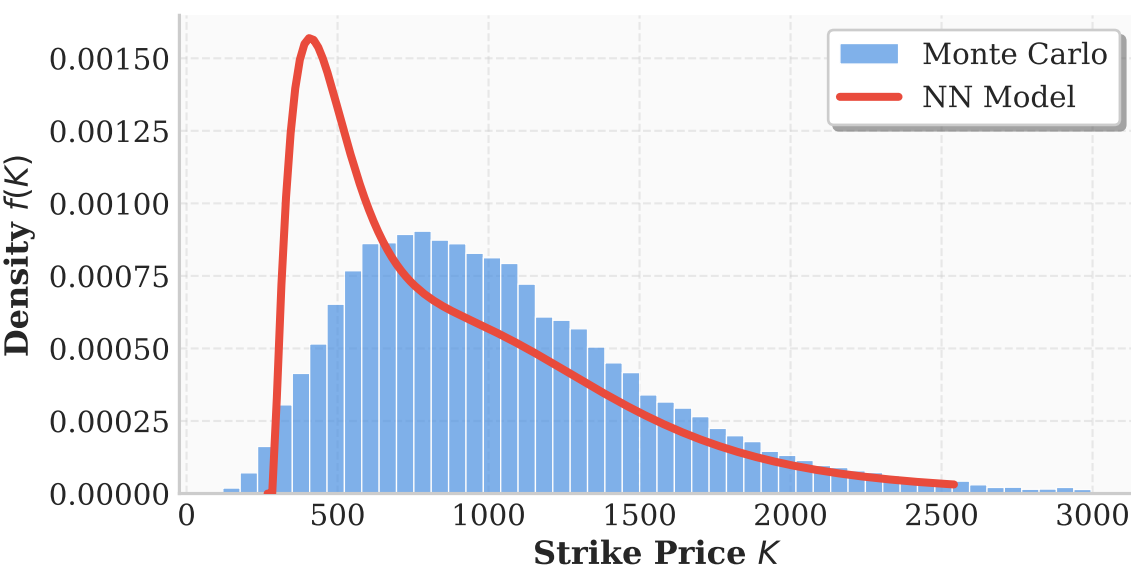
Log-Normal Distribution (T = 1.00)



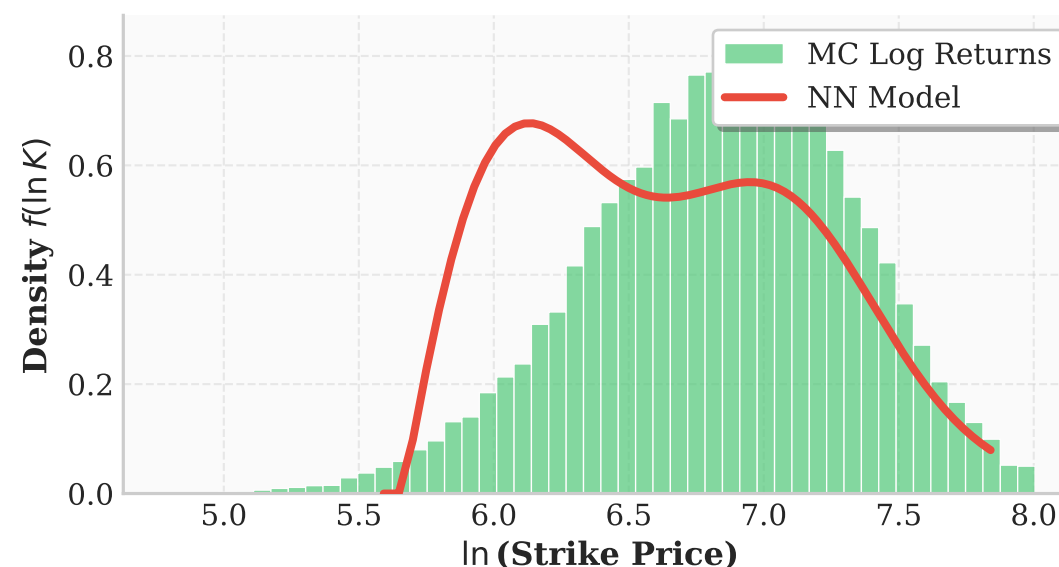
Gaussian Space (T = 1.00)
MC Standardization: $\mu_{MC}=6.87, \sigma_{MC}=0.41 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.69, \sigma_{model}=0.44$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.29, \text{ExKurt}=0.15$



Strike Distribution (T = 1.50)



Log-Normal Distribution (T = 1.50)



Gaussian Space (T = 1.50)
MC Standardization: $\mu_{MC}=6.84, \sigma_{MC}=0.50 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.64, \sigma_{model}=0.51$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.27, \text{ExKurt}=0.09$

